

reloved · cost model

Prepared for Sheetal Ahuja · **By** Totem Interactive · **22 Aug 2026** · Rev 6 — 27 Aug 2026 · Modelled estimates (ranged). Domain + cPanel = confirmed actuals. Totem build fees excluded. Per SOW \$4, third-party charges at approved actuals.

Stack: GoDaddy (domain + cPanel: mail, static frontend) · **Firestore** (backend — API, auth, database, storage; Blaze pay-as-you-go plan) · Brevo (email/OTP) · MSG91 + **TRAI DLT** (SMS OTP) · Gemini (AI cataloguing) · MapTiler (map). No separate VPS.

MAU assumptions (from actual call patterns in the code, not flat guesses): **~2.5 SMS OTP/user/mo** (donor sessions expire in 7 days — confirmed from the session token, so repeat visits mean repeat OTPs) · **~4–5 emails/user/mo** (every donation sends 2: donor confirmation + admin alert; every claim adds 1) · **~0.6 Gemini calls/user/mo** (billed **per photo uploaded**, not per user/donation — up to 5 photos allowed per donation) · **~1 map load/user/mo**. Email OTP needs **no DLT**.

Open item: AI photo processing (Gemini + background removal) has no confirmed home post-VPS. Code currently relays to an external server for native binaries (sharp/ONNX) a plain Cloud Function can't run. Likely fix is **Cloud Run** (same Firebase billing) — not yet built, not priced below.

FIXED COSTS (ANNUAL) & FIREBASE BLAZE

Item	Vendor	Amount
Domain reloved.digital	GoDaddy	₹398/yr ✓
Mail + static frontend hosting	GoDaddy cPanel	₹2,388/yr ✓
Subtotal fixed hosting		₹2,786/yr (~₹232/mo)

Firebase Blaze (Cloud Functions + Firestore + Storage + Hosting) is mandatory pay-as-you-go here — the functions call Brevo/MSG91, and Firebase's free "Spark" plan blocks all outbound networking. No monthly minimum; billed only past the free tier below.

Component	Free tier (no cost within these)	Overage rate
Cloud Functions (1 fn, asia-south1)	2M invocations/mo · 400K GB-s/mo · 200K CPU-s/mo · 5GB out/mo	~₹38/M invocations · compute rate TBC (Tier 2 region, ~10% above US) · ~₹11.45/GB out
Firestore	50K reads/day · 20K writes/day · 20K deletes/day · 1 GiB	~₹5.72/100K reads · ~₹17.17/100K writes · ~₹1.91/100K deletes
Cloud Storage (photos)	5 GB-months (cumulative, not monthly) · 100 GB/mo download	~₹1.90/GB/mo past free tier
Firebase Hosting	10 GB storage · 360 MB/day transfer	₹2.50/GB storage · ₹14/GB transfer

At every modelled MAU (100–10,000), Firestore/Functions/Hosting stay inside the free tier even under generous per-user assumptions (50 reads + 20 writes/user/mo ≈ 16,700 reads/day at 10,000 MAU, vs the 50K/day cap). **Cloud Functions compute time (GB-seconds)** is the one line to actually monitor — photo-relay calls run longer than plain CRUD, and may approach the 400K GB-s/mo free cap at the top end; no confident ₹ figure without real traffic.

INDIA DLT — REQUIRED BEFORE SMS OTP (NOT EMAIL OTP)

Register via **MSG91's linked partner, SmartPing (STPL)** — smartping.live — so the Principal Entity links correctly to MSG91 as Telemarketer. Lead time ~2–7 business days.

Item	Frequency	Amount
Principal Entity (PE) registration	Year 1 + renewal	₹5,900 (₹5,000+GST)
Header (Sender ID) + OTP template	Once	Free
PE → MSG91 telemarketer binding	Once	Free (mandatory)
MSG91 assisted setup (optional)	Once	~₹1,770

Per SMS: MSG91 ₹0.15–0.25 + DLT scrub ₹0.025 → **~₹0.23/SMS all-in. No-DLT launch path:** MSG91's OTP Widget (already integrated, tested working) needs no DLT — use it while DLT registration runs in parallel. Before first raw-template OTP: PE ID · header · template (Service Implicit) · TM 1302157225275643280 chain-bind (*confirm still current*).

USAGE-BASED BY SCALE (MONTHLY, EXCL. FIXED HOSTING & DLT AMORTISATION)

	100 MAU	500	1,000	5,000	10,000
Firebase Blaze — Functions/Firestore/Hosting	PAYG, in-tier*	PAYG, in-tier*	PAYG, in-tier*	PAYG, in-tier*	PAYG, in-tier*
Firebase Storage	PAYG, in-tier**	PAYG, in-tier**	PAYG, in-tier**	PAYG, in-tier**	PAYG, in-tier**
Email (Brevo)	free	free	free	~₹1,800	~₹3,500
SMS/OTP (MSG91+DLT)	₹60–125	₹300–600	₹575–1,150	₹2,875–5,750	₹5,750–11,500
AI photos (Gemini)	~₹4	~₹18	~₹30	~₹150	~₹300
Maps (MapTiler)	free	free	free	~₹2,490	~₹2,490
Usage subtotal	₹64–129	₹318–618	₹605–1,180	₹7,315–10,190	₹12,040–17,790

PAYG = pay-as-you-go — billed automatically past the free tier, no plan upgrade or resizing needed. *"In-tier" means modelled usage at this MAU stays inside the free allowance, so this month's bill is ₹0 — not that the service is free. Cloud Functions compute is the one line to recheck post-launch.

Storage's free allowance is cumulative, not monthly — revisit a few months post-launch regardless of MAU. **Excludes AI photo processing (Gemini/bg-removal) hosting — see open item above; likely a small usage-based Cloud Run line once built.

TOTAL ESTIMATED MONTHLY COST

	100 MAU	500	1,000	5,000	10,000
Fixed hosting	₹232	₹232	₹232	₹232	₹232
DLT PE amortised	₹492	₹492	₹492	₹492	₹492
Usage (above)	₹64–129	₹318–618	₹605–1,180	₹7,315–10,190	₹12,040–17,790
Total — live today	₹788–853	₹1,042–1,342	₹1,329–1,904	₹8,039–10,914	₹12,764–18,514
+ Borzo pickup*	₹3,000	₹3,000	₹3,000	₹3,000	₹3,000

*Not live — ~₹40–60/delivery, ~₹3,000/mo at ~50 deliveries; scales with bookings, not MAU.

EDGE CASES (REAL, CODE-GROUNDED — NOT GENERIC RISK BOILERPLATE)

Scenario	Trigger	Cost/timeline impact
Max-photo donation	App allows up to 5 photos/donation vs ~2 modelled	Gemini calls per donation up to 2.5× modelled — ₹ impact tiny, real risk is hitting free-key rate limits, not cost
Admin bulk-upload burst	Up to 20 items in one commit, each with photos	Gemini + Firestore writes spike in seconds — provision a billed Gemini key before the first bulk run, not the free one
Abandoned donation	Donor's photos get analyzed (step 1) but they never submit (step 5)	Gemini/processing cost is incurred with no live item created — real leakage not visible in the MAU tables above
Single viral-traffic day	Press mention or spike concentrated in one day	Firestore's free tier resets daily (50K reads/day), not monthly — a spike day can exceed it even if the monthly average looks fine; tables above assume even spread
DLT PE rejected/resubmitted	Incomplete application on first pass	No extra cash cost, just delay beyond the 2–7 day estimate — affects only the non-Widget SMS path
FX movement on USD-billed lines	Firebase overage (if any) + future Cloud Run are USD-denominated	All ₹ figures here use today's rate (~₹95.4/\$1); a rupee move shifts those lines only, not the INR-native ones (DLT, MSG91)
Vendor free-tier/pricing changes	Any of Brevo/MapTiler/Gemini/Firebase changes published rates	Every usage row in this doc shifts — recheck live pricing pages before renewal or ~6 months post-launch

Year-one cash (besides usage): ₹2,786 hosting + **₹5,900 DLT PE** (+ optional ₹1,770 MSG91 help) + Firebase Blaze usage (₹0 at modelled scale, no minimum) + **AI processing hosting (not yet priced — see open item)**. Email OTP and Widget phone OTP both launch without DLT; raw-template phone OTP needs DLT first.

Pilot ~₹790–850/mo at 100 MAU (excl. AI processing) · **~₹12,800–18,500/mo** at 10,000 MAU.

Full assumptions: RELOVED_PRICING_FULL.md. Confirm live rates before budget sign-off. GST as applicable.